L7.3: Lengths and angles

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online degree program on data science.

In this video, we are going to talk about the notion of lengths and angles. So, some

of this may have already been seen in Maths 1 in some form and some of this may be really

easy and something you have seen before. But let us recall quickly what is the notion of

lengths and angles.

So, let us talk about the dot product of two vectors in R2. So, consider the vectors, let

us start with an example. So, consider the two vectors, 3, 4 and 2, 7 in R2. So, the

dot product of these two vectors gives us a scalar as follows, scalar is a real number.

So, you have 3 4 dot with 2 7. This is, so you take the first components of each of these

vectors, multiply them. You take the second components of each of these vectors, multiply

them, and you add them. So, this is going to give you 3 times 2 plus 4 times 7. So,

this is 6 plus 28, which is 34.

Let us generalize this to any two vectors in R2.

So, if you have x1, y1 and x2, y2 in R2, the dot product of these two vectors is the scalar

real number computed as follows; x1, y1 dot x2, y2 is x1 times x2 plus y1 times y2. So,

you multiply the first components, you multiply the second components, and then you add those.OK

So, let us now ask what is the length of a vector in R2? So, let us consider 3, 4, which we saw in the previous example while computing dot products. So, if you want to compute the

length of this vector, how do, what do we do? Well, we use the Pythagoras theorem. So,

you drop a perpendicular down to, let us say, the x-axis.

So, then if you do that, you can get, you get a right angled triangle. So, using this

right angled triangle, so this is the right angled triangle. And because this is a right

angled triangle and the, since you have dropped a perpendicular, this length here is 3, this

length here is 4, this is the x coordinate, this part is a y coordinate.

And so if you want to compute the length of the red line, that is the length of the vector,

you use Pythagoras theorem, which says that the length of the vector 3 comma 4 is the

square root of, so the hypotenuse squared is equal to the sum of the squares of the

sides that is a Pythagoras theorem. So, the length of the hypotenuse is square root of

the sum of the squares of the sides.

So, in this case, it is square root of 3 square plus 4 square, square root of 9 plus 16 is

25. So, square root of 25 is 5. And we do not know here what units we are doing it in.

So, whatever units we are doing it in. So, if this one is corresponding to centimeters,

it is in centimeters, if one is corresponding to kilometer, it is in kilometers, if one

is corresponding to miles, it is in miles and so on, so whatever units are being used

for drawing this R2.

So, the units is not important as far as this course is concerned. Of course, if you start

doing physics and so on that matters or if you do real life problems, it matters. So,

the main point here is that you have to remember that this, if you do a general x, y maybe

we can do it right here, if I have x comma y, so if I have something like this, this

you drop a perpendicular here, this represent, this is exactly x and this part is exactly

y. So, it gives you a right-angled triangle. And so the length of this vector, if I call this vector v, so v which is x comma y, its length is square root of x square plus y square.

So, this is really what we are seeing here.

OK. So, observe that, let us link this up with dot products. Observe that 3 comma 4

dot 3 comma 4 is 3 square plus 4 square, which we saw was the exactly what was in the square

root part when we computed length. So, the length of 3 comma 4 is the square root of

the dot product of the vector with itself. So, the length of the vector 3, 4 is square

root of 3 comma 4 dot 3 comma 4, which is square root of 3 square plus 4 square, which

is, this is unfortunately wrong. This is root of 25, which is 5. OK.

So, in general, of course, this is what I just said. The length of the vector is, so

the length of the vector is root of x square plus y square, which is exactly what you get

by doing root of x comma y dot with x comma y and here, as I said, this should be root

25 is 5. So, the length of the vector is given by taking square root of the dot product with

itself. This is the take home. So, the length and the dot product are related in a very

intimate fashion.

Let us look at the angle now. So, the other thing we want to study in this video is angles.

So, the angle between two vectors u and v, so this is a measure of, sorry, this is not,

there is no and here. The angle between the vectors u and v measures how far the direction

is of v from u. So, if you have the angle, this will be a small angle, as compared to

this, which is a large angle. So, in our previous picture or well, so there is a picture coming

up.

So, we have our old friend, 3 comma 4, and we have another vector 1 comma 5. So, the

angle is this thing theta here. So, it measures how far in terms of directions these two vectors

are from each other, so how far the directions of these vectors are from each other. Now,

we have seen this idea of directions and lengths which we called magnitudes in our very first video on vectors, so just recall that if you do not remember.

So, the angle is measured in degrees or radians. So, if it is in degrees, it is between 0 and

360. So, 0 is itself. And as you increase and go up to 360, it comes all the way back

and then 360 is again itself. So, this is like a clock. If you have 9 am and 9 pm, the

clock shows you the same thing, but 12 hours away lab, so that is what this 360 is saying.

If, 360 means you have taken around and come back. But as far as the angle is concerned,

it is 360 and 0 are telling you the same things.

Or it is, it can be measured in terms of radians. So, radians is more of a, it has something

to do with the length of the arc. So, that is measured between 0 and 2 pi. So, again,

0 is like this, meaning they are the same. And 2 pi is when you go all round and come

back. So, just pi which is 2 pi by 2, so half of 2 pi is where they are like this. So, the

two vectors are parallel to each other, but pointing in different directions, like the, from the center of the earth, the North Pole and the South Pole. They are in different

directions.OK So, it is measured in degrees or radians.

So, this also tells you what is the relationship between degrees and radians, because 2 pi

radians is equal to 360 degrees. So, from there you can say, in general, if you have

an angle of theta degrees, what should be the corresponding radians. The angle is often

described by computing or measured, maybe we should say, by computing its trigonometric

functions described or I want to say or measured by computing its trigonometric functions,

so which is sine, cosine, tan, etc.

What is the relation between dot product and angles? So, if you have two vectors in R2,

we can compute the angle between which, let us call it theta, between the two vectors

by using dot products as cosine of theta is u dot v divided by square root of v dot v

times u dot u. So, this times is these are real numbers, remember. So, this is just multiplication in real numbers. So, and then you can take the inverse cosine. And we usually think of

it as being between 0 and 2 pi again.

So, let us do an example. So, consider the two vectors. So, this is in R3. So, now, we

are talking about R3. So, consider the two vectors 1, 2, 3 and 2, 0, 1 in R3. So, the

dot product of these two vectors gives us a scalar as follows. So, 1, 2, 3 dot with

2, 0, 1 is you take component wise multiplication and then you add them. So, 1 times 2 plus

2 times 0 plus 3 times 1, which is 2 plus 0 plus 3, which is 5. So, for two general

vectors, you do the same thing x1, y1, z1, x2, y2, z2 you do x1, y1, z1 dot x2, y2, z2

is x1x2 plus y1y2 plus z1z2.

So, now, let us find the length of the vector, let us say, 4, 3, 3 in R3. So, to do this

we will first draw the xyz plane. So, let us say this is x, y and this z. This is our

right handed coordinate system. Sorry t his is not y, this is y and that is z. So, now, if you take x, y, z, you drop a perpendicular, so this is 4, 3, 3.

So, we will drop a perpendicular down to the xy plane, then join that to the origin, and

then drop further perpendiculars from this point to the x-axis and the y-axis, respectively.

So, what you really have is that this is a perpendicular, this is exactly z. So, in this

case, it is 3. This is 3. And then this is exactly the length of y which is again 3,

the y-coordinate. And this is 4 or, because this is parallel to this, which is 4, this

is parallel to this, which is 3, and this is parallel to this, which is 3.

So, now how do I compute the length? Again, you use Pythagoras theorem. We already know

that this line in the middle here, this line you can compute by Pythagoras theorem as square

root of 3 square plus 4 square, this is root of 5 square which this 5. And then again,

this is the right angle triangle, the green thing, the blue line and the red vector itself.

So, the red, blue and green form of right angle triangle and the red is the hypotenuse, which is exactly the length of the vector 4, 3, 3.

So, length of 4, 3, 3 is root of 5 square, which is the length of the line circled by

the green line, plus 3 square, which is the length of the blue line. But 5 square, I can

write as 3 square plus 4 square, which was the original x and y coordinates. So, this

is 3 square plus 4 square plus 3 square or maybe we should write it as 4 square plus

3 square to keep with the notation, and then you can compute what that is, in this case,

it gives you root 34 units, so 16 plus 9 plus 9.

So, observe again that 4, 3, 3 dot 4, 3, 3 is exactly 4 square plus 3 square plus 3 square.

So, the length of 4, 3, 3 can be expressed as the square root of the dot product of the

vector with itself. This is the same computation. And so the general situation is where you

do x, y, z and the same reasoning tells you that the length of x, y, z is square root

of the dot product of x, y, z with itself. So, we, so this tells you the relationship between the dot product and lengths in, so we have seen the same relation in R2, we have

seen this in R3 and certainly the idea is that this holds for any Rn, the proof is the

same.

Let us now ask for what happens to the angle. So, the angle between two vectors u and v

in R3 is the angle between them computed by passing a plane through them. So, if you have

two vectors u and v, you look at, there is a unique plane passing through those two vectors.

Draw that plane. And now once you are in the plane, it is like you are in R2. And in R2

we know how to compute the angle between two vectors.

And it, so what does this angle do? It measures the direction, how far the direction is of

v from u or u from v on that plane.OK So, we have seen this formula before in R2 and

the same idea holds for R3. So, if u and v are two vectors in R3, we can compute the

angle theta between these vectors by using the dot product as follows. So, cosine of theta is u dot v divided by square root of v dot v times u dot u or you can write thetha

in terms of the inverse cosine of this number that we have just seen.

So, let us maybe try to analyze why this is happening. So, the reason this is happening

is because, so if you have two vectors, so you pass some plane through them, I mean,

there is a unique plane. So, you take this plane. And now on this plane, these vectors

are like two ordinary vectors in R2. So, if you want instead of looking at this plane

you move this plane, rotate it, so that it becomes like the xy plane. And don't disturb

those vectors. When you just do a rotation don't do any scaling or anything else, just

do a rotation.

So, then these vectors are going to look like, on the xy plane they will maybe look like

this. And you can compute the angle between them. And that is exactly what this angle

is. And you can see that this will match with the formula that you have written down. That is the idea.

So, let us compute the angle theta between these two vectors, 1, 0, 0 and 1, 0, 1. If

our intuition is correct, we kind of know what the angle here is. Both of these are

on the plane, the xz plane, y is 0, so xz plane. So, these are on the xz plane. And

on the xz plane, so you just knock out the y coordinate, so on the xz plane, this is

like 1, 0 and 1, 1. So, it is 1, 0 and 1, 1. So, if you think of the xz plane now as

the xy plane this is 1, 0 and 1, 1 and we know how much the angle is. This angle is

exactly 45 degrees or pi by 4 radians, whichever you prefer.

So, let us see if the answer that we get by doing that computation is the same 1, 0, 0

dot 1, 0, 1 is 1, 1, 0, 1 dot 1, 0, 1 is 2, 1, 0, 0 dot 1, 0, 0 is 1. And so, we get theta

is cosine inverse of 1 by root of 2 times 1, which is root 2, which is as we know, is

pi by 4 radians or 45 degrees. So, this last thing is something we are getting from a cosine table. You can look those up.

So, let us look at this second example, the angle between 1, 0, 0 and 1, 1, 1. So, for

1, 0, 0 and 1, 1, 1, let us compute these numbers, 1, 0, 0 dot 1, 1, 1 is 1, 1, 1, 1

dot 1, 1, 1 is 3, which is what is giving that root 3 and 1, 0, 0 dot 1, 0, 0 is 1.

So, we get cosine inverse theta, which is the angle between the two is cosine inverse

of 1 by square root of 1 times 3, which is cosine inverse of 1 by root 3 and you can

look up a table to get what the exact value is. It is something like 54 degrees, if I

remember right.

OK and now we can do the general idea of, make the general definition that is, if you

have two vectors, u and v in Rn, the dot product is defined as component wise multiplication

and then addition, so u1v1 plus u2v2 plus unvn. The length of the vector is, it is denoted

by this two bars. So, it is called a norm, which we are going to study in the next video.

So, it is given by root of u dot product with itself. And the angle between two vectors,

again, you pass a plane through those two vectors, there is a unique plane passing,

and you can then rotate it so that it becomes parallel, it becomes the xy plane and then

you can measure, and it turns out that this is exactly the same as doing cosine theta

is u dot v divided by norm of u times norm of v, which, in other words, is square root

of v dot v times square root of u dot u. And then you can write it in terms of the inverse.

So, we we actually have seen a little bit this before lengths and angles in our video

on vectors. But this is a reminder and it also is a reminder about what are dot products

and how they help in computing the lengths and angles between vectors. Thank you.